MAT 1272 (D078): Statistics

Fall 2026
New York City College of Technology, CUNY
Department of Mathematics

Instructor Information

Instructor:	Prof. Patricia Medina
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Office:	N-727
Class Meetings:	MW, 12:52 PM–2:07 PM
Classroom:	N-702
Office Hours:	MW, 9:45 AM–10:40 AM
Course Platform:	Brightspace

1. Course Description

MAT 1272 is an introduction to statistical methods and statistical inference. Topics include descriptive statistics, random variables, probability distributions, sampling distributions, estimation and inference, hypothesis testing, chi-square tests, and correlation/regression.

This is a compressed four-week summer course. Students should expect a fast pace, daily preparation, frequent practice, and regular attendance.

2. Credits and Prerequisites

Credits: 3

Prerequisite: MAT 1190, MAT 1190CO, or higher. This course is not open to students who have completed MAT 1372 or MAT 2572.

3. Required Materials

Textbook: *Introductory Statistics*, 10th edition, Prem S. Mann, John Wiley & Sons.

Students should have access to the textbook in order to complete suggested practice problems. Homework problems will be suggested regularly, but they will not be collected for a grade.

Calculator: A TI-84 Plus or equivalent graphing calculator is strongly recommended. Calculator technology will be used throughout the course for descriptive statistics, regression, probability distributions, normal distributions, and hypothesis testing.

Brightspace: Brightspace is the official course platform for announcements, course materials, grades, and communication.

4. Course Learning Outcomes

By the end of the course, students will be able to:

1. Define basic statistical terms and distinguish between descriptive and inferential statistics.
2. Organize, construct, and interpret tables and graphs for quantitative and qualitative data.
3. Calculate and interpret measures of central tendency, measures of variation, quartiles, percentiles, and outliers.
4. Use technology to find correlation coefficients, regression lines, and predicted values.
5. Calculate probabilities using probability rules, counting methods, permutations, and combinations.
6. Work with random variables and probability distributions, including binomial and hypergeometric distributions.
7. Use the normal distribution and standard normal distribution to solve applied problems.
8. Apply the Central Limit Theorem and interpret sampling distributions.
9. Conduct basic hypothesis tests using appropriate test procedures.
10. Use chi-square procedures for goodness-of-fit and independence/homogeneity problems.
11. Communicate statistical conclusions clearly and in context.

5. General Education Alignment

This course supports the development of quantitative reasoning, critical thinking, and data literacy. Students will learn to interpret mathematical and statistical information, use appropriate numerical and graphical methods, and draw conclusions from data while recognizing the limits of statistical analysis.

6. Assessment and Grading

Component	**Percentage**
Exam 1	15%
Exam 2	15%
Exam 3	15%
Quizzes	15%
Group Activities / Participation	10%
Comprehensive Final Exam	30%
Total	**100%**

6.1. Quizzes

There will be short quizzes throughout the course, typically one or two per week. Quizzes may be based on lecture material, class examples, suggested textbook problems, and previous class activities.

The lowest two quiz grades will be dropped. No make-up quizzes will be given.

6.2. Group Activities and Participation

Students will participate in collaborative in-class activities designed to reinforce course concepts. These activities may include data analysis, graph interpretation, calculator exercises, probability experiments, and applied statistical reasoning.

The lowest two group activity grades will be dropped. No make-up group activities will be given.

6.3. Homework

Homework will be assigned as suggested practice problems from the textbook. Homework will not be collected for a grade, but it is essential preparation for quizzes, exams, and the final examination.

6.4. Exams

The course includes three in-class exams and one comprehensive final examination. Each in-class exam will be approximately 75 minutes long. The remaining class time on exam days may be used for review or for introducing upcoming topics.

Assessment	**Tentative Date**
Exam 1	Wednesday,September 23rd, 2026
Exam 2	Wednesday, October 21st, 2026
Exam 3	Wednesday, November 23rd, 2026
Comprehensive Final Exam	Monday, December 21st, 2026

Exam dates are tentative and may be adjusted if necessary. Any changes will be announced in class and posted on Brightspace.

7. Grading Scale

A	93–100
A-	90–92.99
B+	87–89.99
B	83–86.99
B-	80–82.99
C+	77–79.99
C	70–76.99
D	60–69.99
F	Below 60

8. Attendance and Lateness

Attendance is expected at every class meeting. This course moves quickly, and missing one class in a four-week summer session is equivalent to missing several classes in a regular semester.

The Mathematics Department attendance policy states that students may be absent without penalty for 10% of scheduled class meetings. Two latenesses, including arriving late or leaving early, count as one absence.

Excessive absences may affect your course grade and may lead to administrative consequences according to college policy.

9. Make-Up Policy

No make-up quizzes or group activities will be given.

Make-up exams are given only in documented cases of valid excused absence, such as illness, jury duty, or emergency circumstances, and are subject to instructor approval. Students should contact the instructor as soon as possible if an emergency occurs.

10. Classroom Environment

This course requires active participation, focus, and mutual respect. Please silence phones and keep distractions to a minimum. We will maintain a classroom environment based on respect, dignity, and support for every student.

Students are expected to participate, ask questions, work collaboratively, and contribute to a serious learning environment.

11. Classroom Recording Policy

To maintain a respectful and supportive learning environment, audio and video recordings are not permitted in this class. Our classroom is a space where everyone should feel comfortable participating, asking questions, and engaging openly with the material. Recording, whether audio or video, can inhibit that openness and disrupt the sense of community we are working to build.

12. Academic Integrity

Academic dishonesty is prohibited at CUNY and New York City College of Technology. This includes cheating, plagiarism, unauthorized collaboration, and submitting work that is not your own.

Penalties for academic dishonesty may include a failing grade on an assignment or exam, a failing grade in the course, suspension, or expulsion. When in doubt, ask before submitting work.

13. Accessibility Statement

The Center for Student Accessibility is the disabilities service provider at New York City College of Technology. The Center provides support and accommodations for students with documented disabilities.

Location:	L-237
Phone:	718-260-5143
Email:	Accessibility@citytech.cuny.edu

Students who need accommodations should contact the Center for Student Accessibility and inform the instructor as early as possible.

14. Communication

Students should check their CUNY email and Brightspace regularly. Announcements, course materials, updates, and grades will be posted on Brightspace.

Do not wait until the night before an exam to ask for help. Students are encouraged to attend office hours and ask questions early.

15. Getting Help

Students are encouraged to:

- attend office hours,
- ask questions during class,
- review class notes regularly,
- complete suggested practice problems,
- use tutoring and academic support resources when available,
- form study groups with classmates.

16. Tentative Course Schedule

This schedule is tentative and may be adjusted depending on class progress. Any changes will be announced in class and posted on Brightspace.

The official course outline contains approximately 30 lessons.

Tentative Course Schedule

The following schedule outlines the topics to be covered throughout the semester. The schedule is tentative and may be adjusted as needed to support student learning.

Meeting	Topics and Textbook Sections
1	Introduction to Statistics; Statistics and Types of Statistics; Basic Terms — Sections 1.1–1.2
2	Types of Variables; Population vs. Sample — Sections 1.3, 1.5
3	Organizing and Graphing Qualitative Data — Section 2.1
4	Organizing and Graphing Quantitative Data — Section 2.2
5	Stem-and-Leaf Displays; Summation Notation — Sections 2.3, 1.7
6	Measures of Central Tendency — Section 3.1
7	Measures of Dispersion; Standard Deviation — Sections 3.2, 3.4
8	Measures of Position; Boxplots and Outliers — Sections 3.5–3.6
9	Simple Linear Regression — Section 13.1
10	Linear Correlation — Section 13.4
11	Experiments, Outcomes, Sample Spaces, and Events — Section 4.1
12	Calculating Probability — Section 4.2
13	Conditional Probability; Independent and Dependent Events — Section 4.3
14	Multiplication Rule; Intersection of Events — Section 4.4
15	Addition Rule; Union of Events — Section 4.5
16	Counting Rules; Factorials; Permutations and Combinations — Section 4.6
17	Hypergeometric Distribution — Section 5.5
18	Random Variables; Discrete Random Variables — Section 5.1
19	Probability Distributions of Discrete Random Variables — Section 5.2
20	Mean and Standard Deviation of a Discrete Random Variable — Section 5.3
21	Binomial Probability Distribution — Section 5.4
22	Continuous Probability Distributions; Normal Distribution — Section 6.1
23	Standard Normal Distribution; Standardizing the Normal Distribution — Section 6.2
24	Applications of the Normal Distribution — Section 6.3
25	Inverse Normal Problems — Section 6.4

Meeting	Topics and Textbook Sections
26	Sampling Distributions; Sampling Error; Non-sampling Error — Section 7.1
27	Mean and Standard Deviation of $\bar{x}$; Sampling Distributions — Section 7.2
28	Central Limit Theorem; Applications of Sampling Distributions — Sections 7.3–7.4
29	Introduction to Hypothesis Testing; Hypothesis Tests for μ — Sections 9.1–9.3
30	Chi-Square Distribution; Goodness-of-Fit; Tests of Independence and Homogeneity — Sections 11.1–11.3

17. Important Notes

- This is a fast-paced summer course. Students should review material daily.
- Suggested textbook problems are not optional in practice: they are the main preparation for quizzes and exams.
- Students are responsible for all announcements made in class and posted on Brightspace.
- The final exam is comprehensive and is worth 30% of the course grade.
- Wednesday, June 24 is reserved for final exam review only.